# Investment Project - Group 1

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# 3 Pulling the data

For the WRDS Monthly World indices and the CRSP's value weighted index for the US we have used WRDS's user interface online to pull the data. For the 1-month T-Bill and the interbank rates we have downloaded the data from the FRED's website.

# 4 The international diversification strategy (DIV)

(a) We are given the market indices of the following countries: Australia, France, Germany, Japan, Switzerland, the United-Kingdom and the United States. The first step consists of computing the dollar returns of these different indices. In order to so, we need exchange rates of the different currencies used for all of these markets. We denote by $R_t^{\text{loc}}$ the return in the local currency of a market index from time $t-1$ to time $t$. We also denote $S_t$ the value in USD of one unit of the local currency at time $t$. The local market return is given by:

$$R_{t+1}^{\text{loc}} = \frac{P_{t+1}^{\text{loc}} + D_{t+1}^{\text{loc}} - P_t^{\text{loc}}}{P_t^{\text{loc}}}$$

Where $P_t^{\text{loc}}$ and $D_t^{\text{loc}}$ are the price of underlying asset and the dividend paid at time $t$ in the local currency. The unhedged return of this index in USD is then given by:

$$R_{t+1}^{\text{unhedged}} = \frac{(P_{t+1}^{\text{loc}} + D_{t+1}^{\text{loc}}) \times S_{t+1} - P_t^{\text{loc}} \times S_t}{P_t^{\text{loc}} \times S_t} = (R_{t+1}^{\text{loc}} + 1) \times \frac{S_{t+1}}{S_t} - 1$$

(b) We then compute the currency hedged stock index returns as:

$$R_t^{\text{hedged}} = R_t^{\text{unhedged}} - X_t$$

where $X_t$ is the excess return in USD to a 1 \$1 investment in the local currency, given by borrowing \$1 at the US 3-months risk free rate $r^{US}$, converting into $\frac{1}{S_t}$ local currency, investing in the 3-months local risk free rate $r^{\text{loc}}$, and converting the proceeds back into dollars after one month at the then prevailing exchange rate $S_{t+1}$:

$$X_{t+1} = \frac{S_{t+1}}{S_t}(1 + r_{t+1}^{\text{loc}}) - (1 + r_{t+1}^{US})$$

(c) We are building three different strategies:

- **Equally weighted** : At each time $t$, we invest the same part of our wealth in each of the risky assets. Wishing to invest all of our wealth, for $N$ different assets, we invest $w_t = \frac{1}{N}$ of our wealth in each asset.

- **Risk Parity** : At each period $t$, we compute the standard deviation of returns of each assets on the 60 past months. The wealth invested in an asset for the period from $t$ to $t+1$ is equal to the inverse of its standard deviation on the previous period, normalized such that the totality of the wealth is equal 1:

$$\sigma_t^i = \text{std}([R_{t-60}^i, \quad ... \quad , R_t^i])$$

$$w_{t+1}^i = \frac{1/\sigma_t^i}{\sum_{j=1}^N 1/\sigma_t^j}$$

  For the end of this paper, we denote by $R^{DIV}$ the returns of the risk parity portfolio built as above.

- **Mean-Variance** : At each time $t$, we compute the covariance of the returns of the $N$ risky assets from $t-60$ to $t$ denoted by $\Sigma_t$, and the mean returns of all assets on the same period denoted by $\mu_t$. With $R$ the vector of risky assets, $R^f$ the risk free return, given a risk aversion $\gamma$, the objective is to build a portfolio $R_{t+1}^P = R_{t+1}^f + w_{t+1}^T(R_{t+1} - R_{t+1}^f \mathbb{1})$ that maximizes the utility function:

$$\begin{aligned} & \max_{\omega_{t+1}} \left\{ \mathbb{E}_t[R_{t+1}^P] - \frac{\gamma}{2}\text{Var}_t(R_{t+1}^P) \right\} \\ \iff \quad & \max_{\omega_{t+1}} \left\{ \omega_{t+1}^\top(\mu_t - R_t^f \mathbb{1}) - \frac{\gamma}{2}\omega_{t+1}^\top \Sigma_t \omega_{t+1} \right\} \end{aligned}$$

  This gives

$$w_{t+1} = (\gamma\Sigma)^{-1}\left(\mu_t - \mathbb{1}R_{t+1}^f\right)$$

We report the results of the three strategies using Currency-hedged returns in in Table 1 an unhedged returns in Table 2. These tables give the annualized mean return, volatility and Sharpe ratio.

| **Portfolio Strategy** | **Mean** | **Standard Deviation** | **Sharpe Ratio** |
|---|---|---|---|
| Equal Weighted | 7.6704% | 13.098% | 0.4724 |
| Risk Parity | 6.5275% | 13.143% | 0.4012 |
| Mean-Variance Optimal ($\gamma = 1$) | -1.8132% | 171.73% | -0.0176 |
| Mean-Variance Optimal ($\gamma = 10$) | 0.8789% | 17.191% | -0.0188 |

TABLE 1: Annualized Performance of Currency-Hedged Portfolio Strategies

| Portfolio Strategy | Mean | Standard Deviation | Sharpe Ratio |
|---|---|---|---|
| Equal Weighted | 8.6558% | 16.584% | 0.4326 |
| Risk Parity | 5.7934% | 16.998% | 0.2670 |
| Mean-Variance Optimal ($\gamma = 1$) | 19.0189% | 192.65% | 0.0923 |
| Mean-Variance Optimal ($\gamma = 10$) | 2.9948% | 19.265% | 0.0912 |

TABLE 2: Performance of Currency-Unhedged Portfolio Strategies

We first computed the mean-variance efficient portfolio using risk aversion $\gamma = 1$. Doing so, the investor would end with a cumulative return of $-100\%$ and thus lost all of his wealth, as displayed on Figure. 4.1

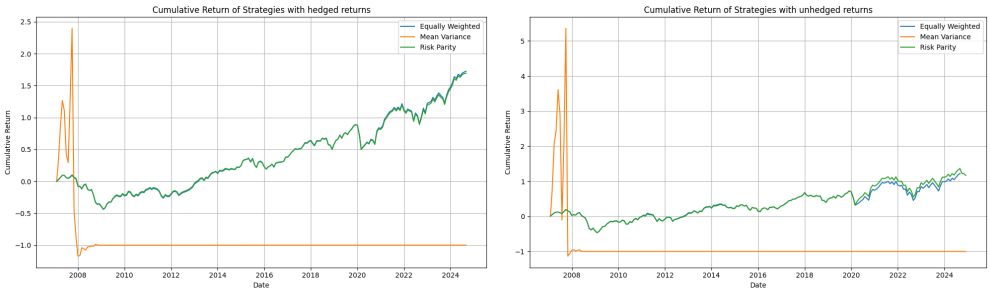


FIG. 4.1: Cumulative returns of the three strategies, taking a risk aversion $\gamma = 1$ to build the MVE

We then computed the strategy for a bigger risk aversion $\gamma = 10$, the cumulative returns are displayed on Figure. 4.2

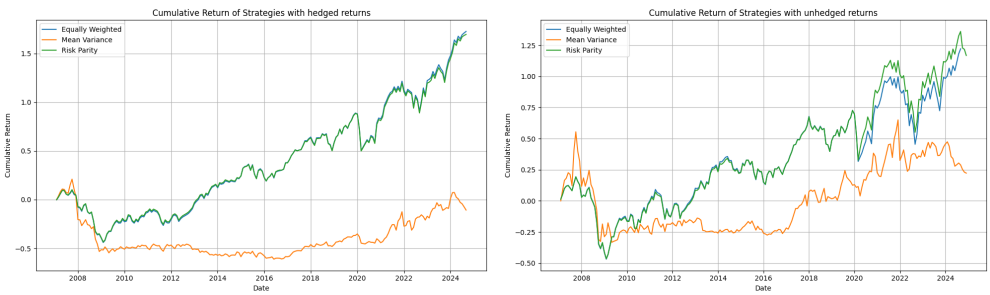


FIG. 4.2: Cumulative returns of the three strategies, taking a risk aversion $\gamma = 10$ to build the MVE

*Conclusion*: From both Table 1 and Table 2 we find that

- Hedged portfolios offer higher Sharpe Ratios for Risk Parity and Equally weighted strategies. This suggests that hedging reduces currency risk and improves risk-adjusted performance
- Additionally, the equal-weighted unhedged strategy outperforms its hedged counterpart. This suggests that even a naïve diversification approach benefits from the additional returns caused by exchange rate changes.
- The mean-variance has the lowest Sharpe ratio, coming from both the lowest mean return and highest volatility. We observe here the limitations of the mean-variance optimization observed in Black-Litterman(1991). In fact this unconstrained optimization leads to unreasonable weights coming from a lack of precision in estimation of sample mean an correlation, and the counter-intuitive impact on weight of the correlation matrix.

Hedging foreign exchange exposure tends to improve risk-adjusted returns by lowering portfolio volatility, especially in simpler strategies like equal-weighted and risk parity. These results suggest that for a US investor, there are benefits in international diversification, except when following a mean-variance approach.

# 5 Equity Index Momentum Strategy (MOM)

(a) At each time $t$, we compute the momentum of the currency hedged stock indexes on the last year as the cumulative returns from $t-12$ to $t-1$. For country $i$, the momentum at time $t$ is thus given by:

$$\mathrm{MOM}_t^i = \prod_{\tau=t-12}^{t-1} (R_\tau^i + 1) - 1$$

We then rank all of the countries momentum such $\mathrm{Rank}_t^i = 1$ if country $i$ has the lowest momentum from $t-12$ to $t-1$. We then assign weights to countries such that:

$$\omega_t^i = Z\left(\mathrm{Rank}_t^i - \frac{N+1}{2}\right)$$

defining such that the sum of positive $w_i^t$ is 1, and the sum of negative $\omega_t^i$ is -1. We thus have:

$$Z = \sum_{k \geqslant \frac{N+1}{2}}^{N} \left(k - \frac{N+1}{2}\right)$$

For the end of this paper, we denote by $R^{\mathrm{MOM}}$ the returns of the portfolio built using these weights.

(b) The results of this strategy are displayed on Table 3. We also built a long only strategy where we only take long positions on assets with positive weights and a short only strategy where we only take short positions on assets with negative weights.

| Portfolio Strategy | Mean | Standard Deviation | Sharpe Ratio |
|---|---|---|---|
| MOM Long Leg | 7.3088% | 13.807% | 0.4240 |
| MOM Short Leg | -9.9817% | 13.288% | -0.8606 |
| MOM Strategy | -2.6729% | 9.0213% | -0.2963 |

TABLE 3: Performance of the Momentum Strategy Components

The cumulative returns of the three strategies are displayed of Figure. 5.1. As one can notice, the short only strategy is overall giving negative returns. This shows the limitation of choosing the weight invested in an asset only based on its momentum rank, and not on the actual value.

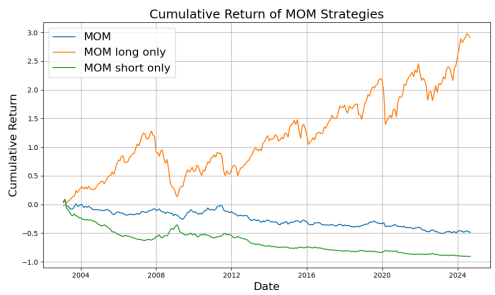


FIG. 5.1: Cumulative returns of momentum strategy

We then test if the MOM strategy has an average return that is statistically significantly different from zero. In order to do so we compute a one sample T-test with the null hypothesis that the average return of the MOM strategy is 0. The t-statistic is $-1.3765$, corresponding to a p-value of 0.1699. We therefore fail to reject the null hypothesis at the 5% significance level, and conclude that the average return of the MOM strategy is not statistically significantly different from zero.

(c) We then run a linear regression using the ordinary least square (OLS) of the MOM strategy return on the DIV return. The results are

- An intercept of approx. $-0.0025$ that is not statistically significant ($p = 0.172$). This suggest no excess return when the DIV return is zero.
- The slope coefficient is 0.0793 and is statistically significant at the 10% level with a p-value of 0.092.
- Finally, we get an $R^2$ of 0.014, so the DIV strategy explains 1.4% of the variation in the MOM returns, which is very low.

*Conclusion* : The low $R^2$ and small slope coefficient suggest that MOM returns are largely independent from DIV returns. This implies that MOM can diversify DIV returns, and thus DIV-investors could use this very low correlation and invest in MOM in order to diversify their portfolios.

*Going further*: The negative return of the strategy on the whole period motivates us to inverse the sign of the weights assigned to every assets. The results are displayed on Figure. 5.2. We notice that this time the strategy has an overall positive cumulative return on the period. This suggests that firms with lowest momentum are more likely to have highest returns in the following period in this international context.

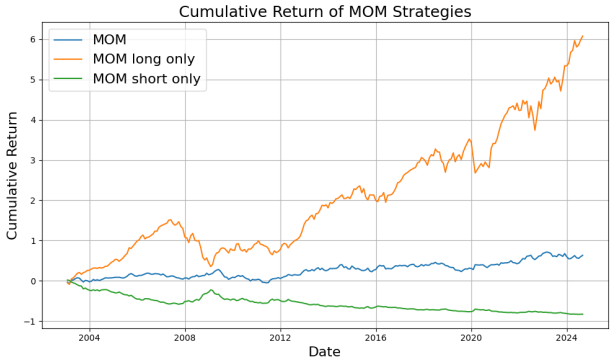


FIG. 5.2: Inverted momentum strategy cumulative returns

# 6 Equity Index Long Term Reversal strategy (REV)

(a) At each time $t$, we compute the long term reversal of the currency hedged stock indexes on the last year as the cumulative returns from $t-60$ to $t-12$. For country $i$, the long term reversal at time $t$ is thus given by:

$$REV_t^i = \prod_{\tau=t-60}^{t-12} (R_\tau^i + 1) - 1$$

We then rank all of the countries long term reversal such that $\text{Rank}_t^i = 1$ if country $i$ has the lowest cumulative returns from $t-60$ to $t-12$. We then assign weights to countries such that:

$$\omega_t^i = Z\left(\frac{N+1}{2} - \text{Rank}_t^i\right)$$

defining such that the sum of positive $w_i^t$ is 1, and the sum of negative $\omega_t^i$ is -1. We thus have:

$$Z = \sum_{k \geqslant \frac{N+1}{2}}^{N} \left(k - \frac{N+1}{2}\right)$$

For the end of this paper, we denote by $R^{REV}$ the returns of the portfolio built using these weights.

(b) Here we give the table with the all the values (i.e. mean, standard deviation and Sharpe Ratio) for the REV long leg, short leg, and full.

| **Portfolio Strategy** | **Mean** | **Standard Deviation** | **Sharpe Ratio** |
|---|---|---|---|
| REV Long Leg | 6.5430% | 13.450% | 0.3971 |
| REV Short Leg | -5.9580% | 14.169% | -0.5053 |
| REV Strategy | 0.5850% | 7.5496% | 0.0775 |

Table 4: Performance of the Reversal Strategy Components

The cumulative returns of the three strategies are displayed on Figure. 6.1. As we can see, the short only was performing well until 2009, but ended with overall negative cumulative return on the period. It is the exact inverse for the long only position. We can see that overall, this zero cost strategy has approximately a zero cumulative return on the whole period.

We then test if the REV strategy has an average return that is statistically significantly different from zero. In order to do so we compute a one sample T-test with the null hypothesis that the average return of the REV strategy is 0. The t-statistic is 0.3249,

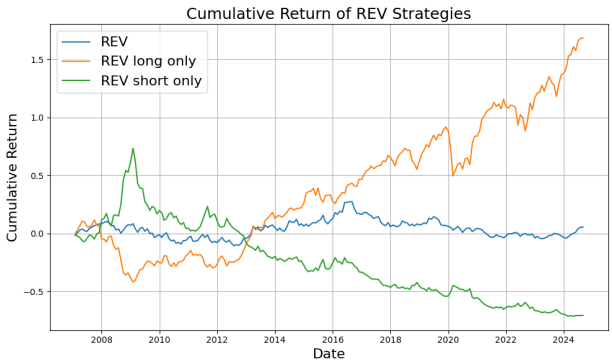


FIG. 6.1: REV strategies cumulative returns

corresponding to a p-value of 0.7456. We therefore fail to reject the null hypothesis at the 5% significance level, and conclude that the average return of the REV strategy is not statistically significantly different from zero.

(c) We then run a linear regression using the ordinary least square (OLS) of the REV strategy return on the DIV return. The results are

- The intercept is approximately 0.0009, but it is not statistically significant ($p = 0.57$). This suggests no consistent excess return when the DIV return is zero.
- The slope coefficient is $-0.068$, and is statistically significant at the 10% level (with a p-value of $p = 0.086$).
- The $R^2$ is very low (0.014), meaning only 1.4% of the variation in REV returns is explained by the DIV returns. This implies a very weak explanatory power.

*Conclusion*: The REV strategy appears to be largely independent from the DIV strategy. For a DIV-investor, this low correlation suggests that REV could offer valuable diversification benefits, even though the strategy itself does not show statistically significant excess returns.

# 7 Currency Carry Strategy (CARRY)

(a) Every month, we sort currency returns based on the difference between the foreign risk-free 3 month interest rate and the 3 months USD risk-free rate. Note that as Germany and France have the same currency (the Euro) and the same interbank rate, we only take into account one of them. We do it such that $\text{Rank}_t^i = 1$ if country $i$ has the lowest interest rate differential at time $t$. We then compute the following weights

$$\omega_t^i = Z\left(\text{Rank}_t^i - \frac{N+1}{2}\right)$$

defining such that the sum of positive $\omega_t^i$ is 1, and the sum of negative $\omega_t^i$ is -1. We thus have:

$$Z = \sum_{k \geqslant \frac{N+1}{2}}^{N} \left(k - \frac{N+1}{2}\right)$$

The return of this strategy is:

$$R_{t+1}^{CARRY} = \sum_{i=1}^{N} \omega_t^i X_{t+1}^i$$

(b) Here we give the table with the all the values (i.e. mean, standard deviation and Sharpe Ratio) for the CARRY long leg, short leg, and full.

| **Portfolio Strategy** | **Mean** | **Standard Deviation** | **Sharpe Ratio** |
|---|---|---|---|
| CARRY Long Leg | 2.2767% | 10.058% | 0.0818 |
| CARRY Short Leg | -0.7198% | 8.2679% | -0.2630 |
| CARRY Strategy | 1.5569% | 9.0994% | 0.1711 |

Table 5: Performance of the Currency Carry Strategy Components

The cumulative returns of these three strategies are plot on Figure. 7.1. The CARRY long-only strategy consistently outperforms both the combined and short-only strategies, achieving a peak return above 100% before 2012 and maintaining a clear upward trend, albeit with some draw-downs in the post-2015 period. In contrast, the CARRY short-only strategy performs poorly throughout the sample, generating persistent negative returns and acting as a clear drag on the overall strategy. The CARRY combined strategy, which presumably takes both long and short positions, delivers positive but modest cumulative returns. This once again highlights the limitation of assigning portfolio weights strictly based on the rank of assets according to a criterion.

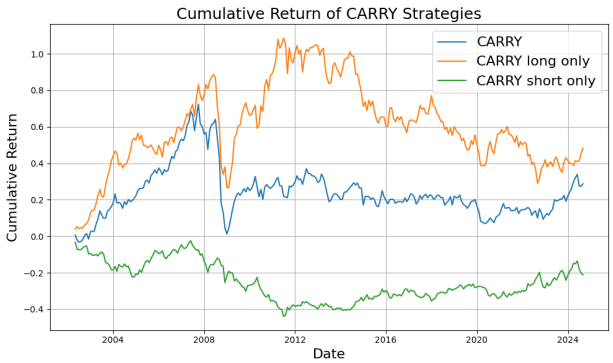


FIG. 7.1: Carry strategies cumulative returns

We then test if the CARRY strategy has an average return that is statistically significantly different from zero. In order to do so we compute a one sample T-test with the null hypothesis that the average return of the CARRY strategy is 0. The t-statistic is 0.8086, corresponding to a p-value of 0.4195. We therefore fail to reject the null hypothesis at the 5% significance level, and conclude that the average return of the CARRY strategy is not statistically significantly different from zero.

(c) We then run a linear regression using the ordinary least square (OLS) of the CARRY strategy return on the DIV return. The results are

- The intercept is approximately $-0.0025$ and is not statistically significant ($p = 0.119$). This suggests no conxistent excess return when the DIV return is 0.
- The slope coefficient is 0.3888 and is highly significant with a $p$-value effectively equal to zero. This indicates a positive relationship between the CARRY and DIV strategies. When the DIV return increases by 1 unit, the CARRY return tends to increase by about 0.39 units on average.
- The $R^2$ is 0.285, meaning the DIV strategy explains 28.5% of the variation in the CARRY strategy's returns, which is pretty high compared to the previous regressions.

*Conclusion*: Since the CARRY strategy is correlated with the DIV strategy and a substantial portion of its variation is explained by it, a DIV-investor may already be capturing a significant part of the CARRY exposure. Therefore, adding the CARRY strategy to a DIV-based portfolio may offer limited diversification benefits unless the investor specifically seeks to increase exposure to the same return drivers.

# 8 Currency dollar Strategy (DOLLAR)

(a) We construct a portfolio that goes short an equal weighted basket of all the foreign currencies against the dollar such that:

$$R_{t+1}^{\text{DOLLAR}} = -\frac{1}{N}\sum_{i=1}^{N} X_{t+1}^{i}$$

(b) Here we give the table with the all the values (i.e. mean, standard deviation and Sharpe Ratio) for the DOLLAR return strategy.

| **Portfolio Strategy** | **Mean** | **Standard Deviation** | **Sharpe Ratio** |
|---|---|---|---|
| DOLLAR Strategy | -1.1570% | 7.6442% | -0.1514 |

TABLE 6: Performance of the Currency Dollar Strategy

We then test if the DOLLAR strategy has an average return that is statistically significantly different from zero. In order to do so we compute a one sample T-test with the null hypothesis that the average return of the DOLLAR strategy is 0. The t-statistic is $-0.7193$, corresponding to a p-value of 0.4726. We therefore fail to reject the null hypothesis at the 5% significance level, and conclude that the average return of the DOLLAR strategy is not statistically significantly different from zero. We also display the cumulative return of the strategy on Figure. 8.1

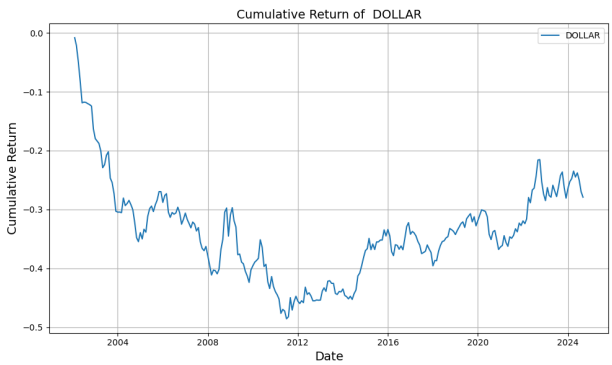


FIG. 8.1: *DOLLAR* strategy cumulative returns over time

(c) We then run a linear regression using the ordinary least square (OLS) of the DOLLAR strategy return on the DIV return. The results are

- The intercept is approximately 0.0015 and is not statistically significant ($p = 0.303$), suggesting that there is no excess return when the DIV return is zero.
- The slope coefficient is $-0.1986$ and is highly significant with a $p$-value effectively equal to zero. This indicates a statistically significant positive correlation between the DOLLAR and DIV strategies.
- The $R^2$ is 0.115, meaning that the DIV strategy explains 11.5% of the variation in the DOLLAR strategy's returns.

*Conclusion*: The regression analysis suggests a statistically significant negative relationship between the DOLLAR and DIV strategies, with a slope around $-0.20$. This suggests that the DOLLAR strategy performs poorly when the DIV strategy performs well, and conversely. Though the explanatory power ($R^2 = 11.5\%$) is poor, the negative relationship is an indication of possible diversification benefits. Thus, a DIV-investor would be interested in the DOLLAR strategy not to add exposure to the same drivers of return, but to hedge against periods of decline in the DIV strategy, and thus improve the overall risk-return profile of his or her portfolio.

# 9 Optimal Fund Portfolio Return (STRAT)

1. A fund investing in the 1-month T-Bill and the diversified stock-index DIV would have return of the form $R_{FUND} = R_{T-Bill} + a(R_{DIV} - R_{T-Bill})$. Suppose that an investor is aiming for an annualized volatility $\sigma_{FUND} = 15\%$, the fund needs to compute $a$ in order to do so. When building a strategy, only the past data is available, and we can not just chose $a$ such that the volatility on the full period is 15% and say that this is the strategy of the fund. We decide to compute $a$ on rolling windows of 60 months of data, such that the volatility on the past 5 years of data would be 15%. We will thus chose $a_t$, for the period $t \to t+1$, such that $\text{STD}(R_{FUND}) = \sigma_{FUND}$ if we took $a = a^t$ on the window $[t-60, t]$. We note $\sigma^t_{DIV} = \text{STD}([R^\tau_{DIV}]_{\tau \in [t-60,t]})$. Since we are implementing a rolling window strategy, changing weights every month, the $T-Bill$ return is known at the beginning of each period and do not generate any volatility in the strategy returns. The volatility of the fund returns on the window $[t-60, t]$, taking $a = a^t$ is thus given by $\sigma^t_{FUND} = a^t \sigma^t_{DIV}$. Finally, since we want this to be equal $\sigma_{FUND} = 15\%$ when annualized, we have:

$$a^t = \frac{\sigma_{FUND}}{\sqrt{12} \times \sigma^t_{DIV}}$$

The time series of $a^t$ is showed on Figure. 9.1. The average value is $\bar{a} = 1.27$.

FIG. 9.1: Value of $a$ through time to get an annualized volatility of 15%

2. We then build a risk parity based portfolio with the four strategies $MOM, REV, CARRY$ and $DOLLAR$. We call the returns of this strategy $R_{STRAT}$, this is done the exact same way as it has been done before, and the cumulative returns of the strategy are displayed on Figure. 9.2. The evolution of weights across time is also displayed on Figure. 9.3.The $REV$ strategy holds the highest weight at both the start and end of the period, while the $DOLLAR$ strategy receives the largest weight for the majority of the timeframe. Although $MOM$ begins with the second highest weight, it rapidly declines to become the lowest-weighted strategy, a position it maintains throughout the remainder of the period. The performance of this strategy is also given in Table. 7

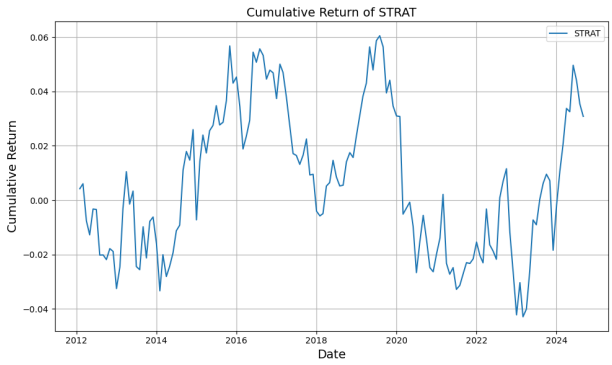


FIG. 9.2: STRAT cumulative returns across time

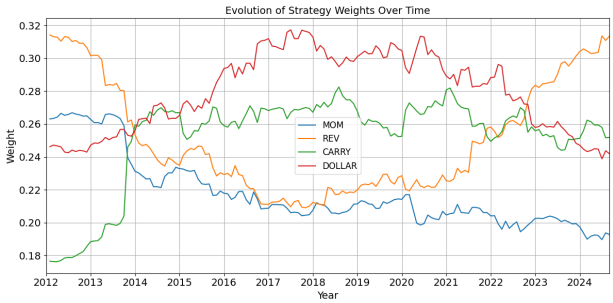


FIG. 9.3: risk parity weights of STRAT across time

| **Portfolio Strategy** | **Mean** | **Standard Deviation** | **Sharpe Ratio** |
|---|---|---|---|
| STRAT Strategy | 0.3146% | 13.279% | 0.082 |

TABLE 7: Performance of the STRAT rolling risk parity Strategy

3. We're looking to build a new portfolio $R_{FUND} = R_{T-Bill} + b \times (R_{DIV} - R_{T-Bill}) + c \times R_{STRAT}$. We're aiming an annualized volatility $\sigma_{FUND} = 15\%$ once again. To do so, we consider the two risky assets $R_{DIV} - R_{T-Bill}$ and $R_{STRAT}$ only source of volatility. We denote by $\Sigma_t$ the covariance matrix of $R_{STRAT}$ and $R_{DIV} - R_{T-Bill}$, and $\mu_t$ the vector of their expected returns on the window $[t-60, t]$. We have the following optimization problem on every rolling windows:

$$max_{w_t} \quad w_t^T \mu_t \qquad \text{such that} \qquad \sqrt{w_t^T \Sigma_t w_t \times 12} = \sigma_{FUND}^t$$

or equivalently:

$$max_{w_t} \quad w_t^T \mu_t \qquad \text{such that} \qquad w_t^T \Sigma_t w_t = \frac{(\sigma_{FUND}^t)^2}{12}$$

The solution is

$$w_t = \frac{\sigma^t_{FUND} \times \Sigma_t^{-1}\mu_t}{\sqrt{12 \times \mu_t^T \Sigma_t^{-1}\mu_t}}$$

where $w_t$ is a vector with components $b_t$ and $c_t$, representing the weights we decide to invest in $R_{DIV} - R_{T-Bill}$ and $R_{STRAT}$ on the period $t \to t+1$. The time series of $b^t$ and $c^t$ are displayed on Figure. 9.4, we notice that $b^t$ doesn't fluctuate as much as $c^t$, indicating that $R_{DIV}$ is less volatile than $R_{STRAT}$. On average, we have $\bar{b} = 1.26$ and $\bar{c} = -0.79$. We notice also the impact of COVID on $c^t$ around 2020-2021, while it is not noticeable on $b^t$.

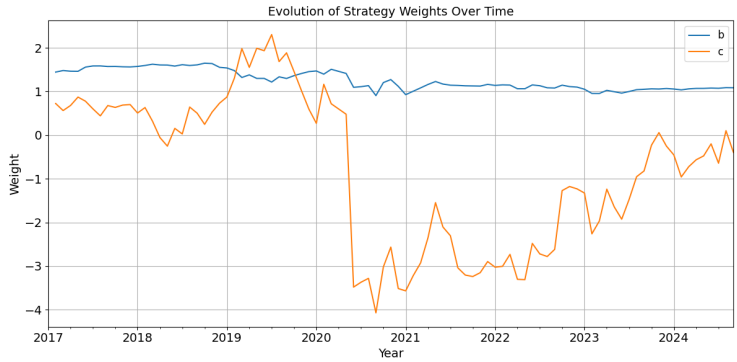


FIG. 9.4: Evolution over time of $b^t$ and $c^t$ when aiming $\sigma_{FUND} = 15\%$ on rolling windows

The time series of the weights invested in each strategies and assets are displayed on Figure. 9.5, with the average values reported in Table. 8. The highest weight is allocated to the *DIV* strategy. In contrast, the weight assigned to *T-Bill* is predominantly negative, indicating that investors would borrow against risk-free assets to achieve a target volatility of 15%. The weights across the various strategies dropped sharply during the COVID period—often turning negative—before gradually recovering to approximately zero by the end of the period.

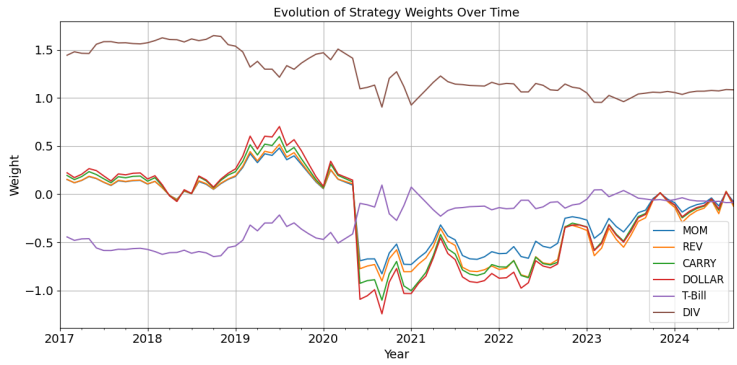


FIG. 9.5: FUND strategy weights time series

| Substrategy | Average Optimal Weight |
|---|---|
| T-Bill | -0.260 |
| DIV | 1.260 |
| MOM | -0.158 |
| REV | -0.210 |
| CARRY | -0.209 |
| DOLLAR | -0.218 |

TABLE 8: Optimal Weights for Each Substrategy in the Fund

4. Now that all the weights have been computed for the two strategies: investing in $DIV$ and $T - Bill$ only versus investing in $STRAT$ additionally, we can compute the returns on the whole period. We first give the performance of both strategies in Table.9. We also plot the cumulative returns of investing $1 in each of the strategies, these are plots on Figure. 9.6. We notice that we have better results when not investing in the $STRAT$, which is against our expectations. It has both a larger expected return and a lower volatility, outperforming the Overlay strategy on all points.

| Strategy | Mean | Standard Deviation | Sharpe Ratio |
|---|---|---|---|
| T-Bill + DIV (Simple Fund) | 11.315% | 16.471% | 0.6767 |
| Overlay Strategy Fund | 9.3516% | 18.243% | 0.5033 |

TABLE 9: Performance Comparison of Simple and Overlay Fund Strategies

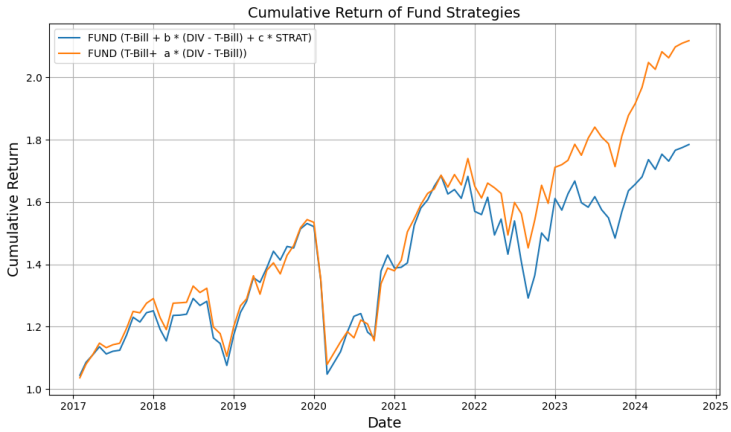


FIG. 9.6: Graph of the cumulative performance of $1 invested in both strategies

# 10 Performance and risk analysis for the Fund strategy

a) We regress the returns of our new strategy on the Fama-French 5 research factors. The results can be found in the following table:

| **Factor** | **Coefficient** | **t-stat** |
|---|---|---|
| $\alpha$ | $-0.0009$ | $-0.195$ |
| Mkt–RF | 0.9032 | 12.366 |
| SMB | 0.1396 | 1.020 |
| HML | 0.0661 | 0.549 |
| RMW | 0.2032 | 1.219 |
| CMA | 0.1052 | 0.596 |
| RF | $-0.0815$ | $-0.04$ |

Table 10: Regression Coefficients and t-statistics

What we have found is that only the market factor (MKT - RF) has a t-stat above $|1.96|$ so unconditionally our strategy returns load significantly on the US-Market. On the other hand we find that none of the other 4 factors (SMB, HML, RMW, CMA) are statistically significant.
The $R^2 = 0.705$ meaning that only about 71% of the variation in our strategy can be explained by the factors used in Fama-French 5. Also the $\alpha = -0.09\%$ is not statistically significant (t $= -0.195$) meaning that we cannot reject the null hypothesis $\alpha = 0$

b) In this subsection we study how our model is consistent with various asset pricing theories.

- **Efficient Markets** : In an efficient market the key implication for a regression of returns is that we should not be able to find any predictable patterns that could generate abnormal profits after we adjust for risk.
  We have found that we could not reject the null that $\alpha = 0$ so we conclude that there is no clear "free lunch" once we control for market risk. This is consistent with he weak form of EMH.
- **CAPM** : When regressing on the Fama-french 5 factors we have seen that the only statistically significant coefficient is the one of the market meaning that our strategy seems to be driven by systematic risk alone. This is exactly in accordance with what CAPM predicts. In addition, the $R^2$ measure of 0.705 is very, which implies that the model explains over 70% of variability in the return of our strategy. This confirms the idea that the performance of our strategy is largely due to systematic risk, and CAPM gives a reasonably good fit in this case.

- **APT**: APT holds that returns are explained by several factors not only the market portfolio. Out of the five factors only the market factor is significant because the others have low betas and low t-stats. However, the moderately strong $R^2$ of 70.5% suggests that a very high percentage of our strategy's return variability is accounted for by the five factors we're concerned with. This leaves open the possibility that APT remains compatible with our findings—it's just that much of the explanatory power is due to market exposure. We can't eliminate APT based on this regression alone, but further testing with other or additional factors may be needed to affirm it.